EXPERIMENTAL MATHEMATICS

Noncorrelation Is a Certificate, Not a Monte Carlo Guarantee

Automatic sequences, exact experiments, and Sobol digital shifts

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THE QUESTION

Can a deterministic sequence with zero lag correlations replace randomness in integration?

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Which deterministic stream would you trust?

OPENING TEST | COMMIT BEFORE SEEING THE CERTIFICATE

A. Thue–Morse

+	−	−	+	−	+	+	−
−	+	+	−	+	−	−	+
−	+	+	−	+	−	−	+
+	−	−	+	−	+	+	−

B. Rudin–Shapiro

+	+	+	−	+	+	−	+
+	+	+	−	−	−	+	−
+	+	+	−	+	+	−	+
−	−	−	+	+	+	−	+

Audience vote: A, B, or "the prefix is not enough information"?

Which deterministic stream would you trust?

OPENING TEST | COMMIT BEFORE SEEING THE CERTIFICATE

A. Thue–Morse

$$\begin{array}{cccccccc} + & - & - & + & - & + & + & - \\ - & + & + & - & + & - & - & + \\ - & + & + & - & + & - & - & + \\ + & - & - & + & - & + & + & - \end{array}$$

B. Rudin–Shapiro

$$\begin{array}{cccccccc} + & + & + & - & + & + & - & + \\ + & + & + & - & - & - & + & - \\ + & + & + & - & + & + & - & + \\ - & - & - & + & + & + & - & + \end{array}$$

B is NC; A is not. Yet this still does not identify the better integration rule.

NC means orthogonality at every fixed nonzero lag

For a finite binary pattern set A,

$$a_{A,k}(n) = \omega_k^{\#(A,n)}, \qquad \omega_k = e^{2\pi i/k}.$$

At $k = 2$, $a(n) \in \{-1, +1\}$. Define

$$\gamma_N(h) = \frac{1}{N} \sum_{n<N} a(n)\overline{a(n+h)}.$$

Noncorrelated: $\gamma(h) = \lim_{N\to\infty} \gamma_N(h) = 0$ for every fixed $h \neq 0$.

$$\gamma(h) = 1$$

perfect agreement with the shifted copy

$$\gamma(h) = 0$$

asymptotic balance at lag h

$$G_N$$

maximum tested correlation, $1 \le h \le 64$

Source: Final report, Sections 1.1–1.2

Automaticity makes exact computation finite

Finite-state proposition. Every pattern-counting sequence $a_{A,k}$ is 2-automatic. If L is the longest word, the automaton stores only

- the last $L-1$ binary digits;
- the accumulated count modulo k.

Hence at most $2^{L-1}k$ states determine the next value.

2
kernel states for Thue–Morse

4
kernel states for Rudin–Shapiro

28
kernel states for the tested A_5 family

Finite self-similarity enables exact recursion, but state complexity alone is not an NC certificate.

Correlation is a residue Fourier coefficient

For a fixed lag, record the count difference

$$D_h(n) \equiv \#(A, n) - \#(A, n+h) \pmod{k}$$

and let $p_{h,r}^{(N)}$ be the frequency of residue r. Then

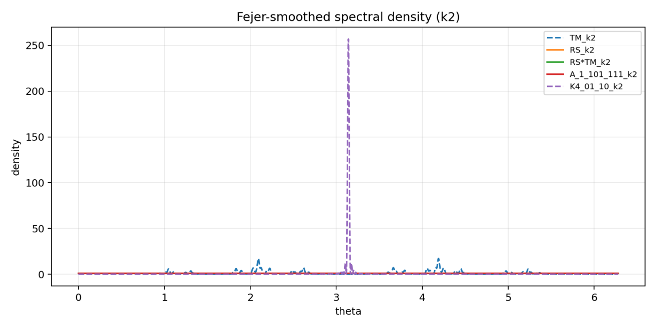
$$\boxed{\gamma_N(h) = \sum_{r=0}^{k-1} p_{h,r}^{(N)} \omega_k^r.}$$

$k = 2$: **parity balance**
NC requires even and odd count differences to balance.

$k = 4$: **two distinct balances**
Parity balance does not force balance within each parity class.

The analytic cancellation problem becomes a finite-state counting problem.

A flat autocorrelation spectrum is equivalent to NC



By Herglotz,

$$\gamma(h) = \int e^{ih\theta}\, d\mu(\theta).$$

For $|a(n)| = 1$,

$$\mathrm{NC} \iff \mu = \lambda/(2\pi),$$

the normalized Lebesgue measure.

1.000

peak-to-mean ratio for all 25 NC catalog cases

41.84

mean ratio for 157 non-NC cases

Rudin–Shapiro is NC by dyadic contraction

Let $r(n) = (-1)^{\#(\{11\},n)}$. Appending one binary digit gives

$$r(2n) = r(n), \qquad r(2n+1) = (-1)^n r(n).$$

Introduce ordinary and mixed correlations

$$\eta_N(h) = \frac{1}{N}\sum_{n<N} r(n)r(n+h), \qquad \vartheta_N(h) = \frac{1}{N}\sum_{n<N} (-1)^n r(n)r(n+h).$$

Even–odd splitting reduces lag $2m$ to m, and lag $2m+1$ to m and $m+1$.

Theorem. $\eta(0) = 1$ and $\eta(h) = 0$ for every $h \neq 0$. The base case contracts at $h = 1$; strong induction closes all higher lags.

Source: Final report, Theorem 3.1; full recurrence in Backup B1

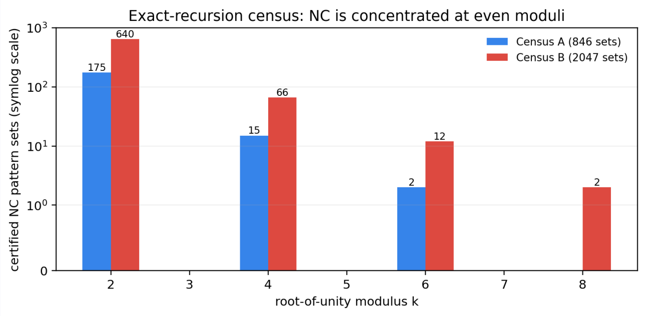
Four experiments test the NC certificate

1. **Exact-recursion censuses:** enumerate pattern sets, moduli 2–8, and lags 1–64.
2. **Spectral cross-check:** compare exact Fejér profiles and statistical summaries.
3. **Prefix convergence:** independently generate $N = 2^{10}, \ldots, 2^{18}$ and fit $G_N \sim N^{-\alpha_G}$.
4. **Closure stress test:** apply ten operations across 1400 catalog rows.

Agreement across exact recursion, spectral flatness, direct prefixes, and transformations is stronger than any single diagnostic.

NC appears almost only at even moduli

EXPERIMENT 1 | 846 + 2047 PATTERN SETS



$$175/640$$

$k = 2$: censuses A / B

$$15/66$$

$k = 4$: censuses A / B

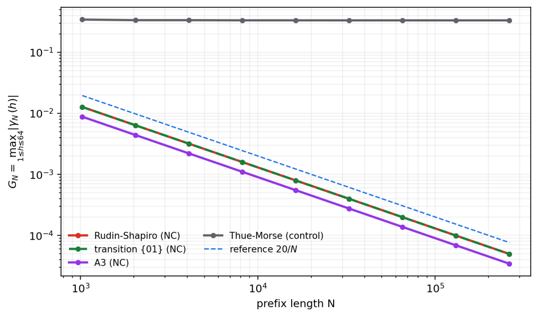
$$0$$

cases at odd $k = 3, 5, 7$; rare cases occur at $k = 6, 8$

This is a finite-universe discovery, not a classification theorem for all pattern sets.

NC prefixes follow an empirical $1/N$ law

EXPERIMENT 3 | INDEPENDENT DIRECT-PREFIX CHECK



$$10$$

certified binary NC examples

$$\bar{\alpha}_G = 1.000$$

mean fitted NC exponent

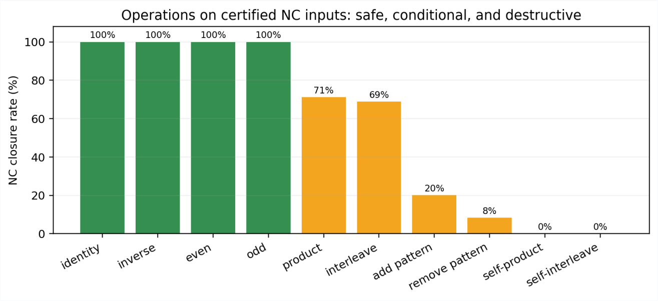
$$\bar{\alpha}_G = 0.0023$$

16 non-NC controls

Tested NC correlation sums remain bounded; non-NC controls converge to a nonzero plateau.

NC is fragile under nonlinear reuse

EXPERIMENT 4 | 1400 CLOSURE TESTS



Always preserved

identity, inverse, even, odd

Conditional

distinct product 71%

distinct interleave 69%

Destroyed

self-product 0%

self-interleave 0%

Vanishing marginal Fourier modes do not force the joint modes needed after combining streams.

Pairwise balance is not multidimensional balance

What NC controls

All second-order lag moments of one source sequence.

What integration needs

Uniform occupancy of many d-dimensional boxes, equivalently much richer Walsh information.

Minimal counterexample

Let X, Y be independent signs. The triple

$$(X, Y, XY)$$

is pairwise independent, but

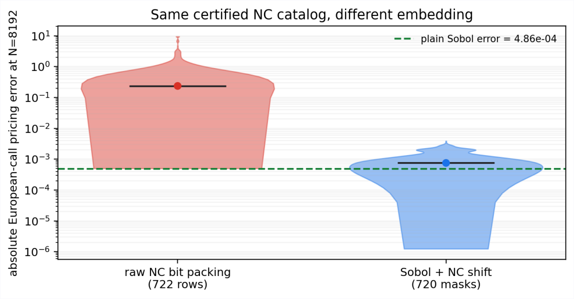
$$X \cdot Y \cdot XY \equiv 1.$$

Pairwise cancellation cannot certify block uniformity or low discrepancy.

Source: Final report, Section 3.7

Embedding changes error by three orders

EXPERIMENT 6 | SAME CERTIFIED NC CATALOG, EUROPEAN CALL



722

raw NC packing rows

0.2362

median raw error

720

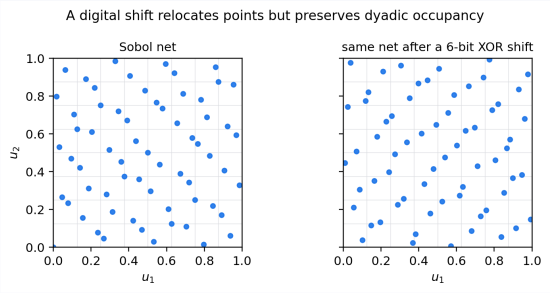
Sobol + NC masks

7.55×10^{-4}

median hybrid error

The NC certificate is unchanged; only the construction of the point set changes.

Digital shifts preserve Sobol nets



Digital-shift theorem

If P is a binary (t, m, s)-net, then

$$P \oplus \Delta$$

is a (t, m, s)-net for every fixed binary mask Δ.

The xor mask permutes dyadic cells without changing their point counts.

The theorem requires a Sobol net and a fixed mask; it does not require the mask to be NC.

Pairing isolates the NC-specific shift effect

1. One identical unscrambled Sobol net per problem.
2. Nested $N = 512, \ldots, 8192$.
3. 64 masks for every family.
4. Eight analytic or symmetry-based references.
5. RMSE across the shift ensemble.

NC families

Rudin–Shapiro, transition, A_3, A_4

Controls

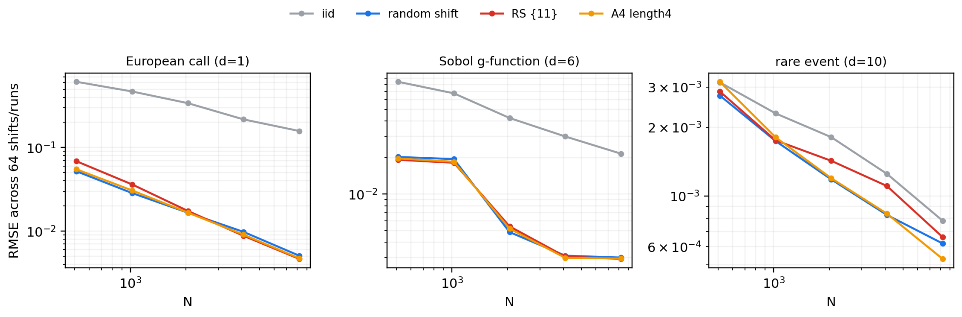
uniform random masks,
Thue–Morse, unshifted Sobol, IID

Deterministic-family RMSE summarizes a fixed mask list; it is not a probabilistic confidence interval.

Pairing removes the main design imbalance from the exploratory benchmark.

NC and ordinary shifts share QMC convergence

EXPERIMENT 8 | PAIRED RMSE CURVES



0.505

mean IID exponent

0.791

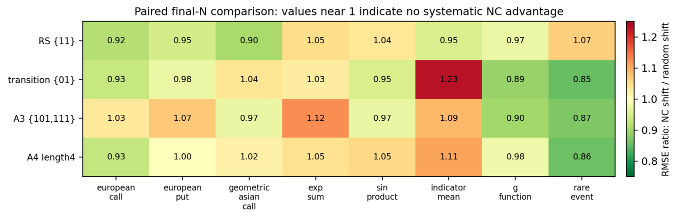
uniform-shift exponent

0.809

Rudin–Shapiro exponent

Source: Focused paired benchmark; eight problems and five nested sample sizes

The controlled comparison finds practical parity



$18/32$

cells favoring NC

0.9894

geometric-mean NC/random ratio

0.9788

median ratio

NC masks are reproducible candidates, but the tested problems show no systematic NC-specific advantage.

The certificate is real; the guarantee is not

1. **NC theory is rigorous.** Automaticity makes correlations finite-state; Rudin–Shapiro is NC by a complete dyadic recursion proof.
2. **The NC experiments are coherent.** Exact censuses, spectral flatness, $1/N$ prefix decay, and closure tests agree on the tested catalog.
3. **Monte Carlo needs more structure.** Raw packing fails; the Sobol backbone supplies multidimensional balance.
4. **NC masks are not universal winners.** Under equal pairing, NC and uniform digital shifts are practically tied.

The next mathematical target is higher-order Walsh structure, not another two-point statistic.

Backup: the full Rudin–Shapiro correlation recursion

PROOF DETAIL

For limiting ordinary and mixed correlations,

$$\begin{aligned}
\eta(2m) &= \tfrac{1}{2}\big(1+(-1)^m\big)\eta(m),\\
\vartheta(2m) &= \tfrac{1}{2}\big(1-(-1)^m\big)\eta(m),\\
\eta(2m+1) &= \tfrac{1}{2}\big((-1)^m\vartheta(m)+\vartheta(m+1)\big),\\
\vartheta(2m+1) &= \tfrac{1}{2}\big((-1)^m\vartheta(m)-\vartheta(m+1)\big).
\end{aligned}$$

At $h=1$, $\vartheta(1)=-\vartheta(1)/2$, so $\vartheta(1)=\eta(1)=0$. Strong induction then reduces every positive lag to smaller ones.

The mixed term is not optional: odd lags couple ordinary and parity-weighted correlations.

Backup: the operational census is not the definition

Mathematical NC

$\gamma(h) = 0$ for every fixed $h \neq 0$ in the infinite limit.

Census label

$\max_{1 \le h \le 64} |\gamma(h)| \le 0.005$ from the specified state recursion.

Independent checks

direct prefixes through 2^{18}; exact Fejér flatness; catalog separation.

Threats

finite pattern universe, padding convention, lag cutoff, floating-point solve.

The project reports the classifier explicitly and does not promote it to a universal theorem.

Backup: aggregate paired benchmark results

EIGHT PROBLEMS | FIVE SAMPLE SIZES

Method	Mean exponent	Median exponent	Geom. final RMSE
IID	0.505	0.507	0.02072
Unshifted Sobol	0.849	0.891	0.001883
Uniform random shift	0.791	0.821	0.001680
Rudin–Shapiro shift	0.809	0.850	0.001645
Transition shift	0.835	0.906	0.001648
A_3 shift	0.812	0.903	0.001680
A_4 shift	0.797	0.840	0.001674
Thue–Morse control	0.835	0.907	0.001749

The non-NC Thue–Morse mask also retains QMC behavior because the Sobol net survives the shift.

Backup: separate theorem, experiment, and inference

LOGICAL STATUS OF THE MAIN CLAIMS

Claim	Status	Reason
Rudin–Shapiro is NC	Theorem	Complete even–odd recursion
Fixed masks preserve Sobol nets	Theorem	Dyadic interval permutation
NC catalog has a $1/N$ prefix law	Finite experiment	Tested through 2^{18}
NC implies low raw discrepancy	Contradicted here	Pair moments miss box frequencies
NC masks beat uniform shifts	Not supported	Ratio 0.9894
Deterministic mask spread is a confidence interval	False without a model	No probability law on the fixed list